Paleomagnetism of the Triassic Nikolai Greenstone, McCarthy Quadrangle, Alaska

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Paleomagnetic evidence indicates that the extensive early Mesozoic basalt field near McCarthy, south-central Alaska, originated far south of its present position relative to North America. Results obtained from the Middle and (or) Upper Triassic Nikolai Greenstone suggest that those basalts originated within 15° of the paleoequator. This position is at least 27° (3000 km) south of the Upper Triassic latitude predicted for McCarthy on the basis of paleomagnetic data from continental North America. The Nikolai pole, as determined from 50 flows sampled at 5 sites, is at 2.2° N, 146.1° E (α_{95} = 4.8°). The polarity of the pole is ambiguous, because the corresponding magnetic direction has a low inclination and a westerly declination. Therefore, the Nikolai may have originated near 15° N latitude or, alternatively, as far south as 15° S latitude. In addition to being displaced northward, the Nikolai block has been rotated roughly 90° about the vertical axis. A measure of the reliability of this pole is provided by favorable results from the following tests: (1) Within one stratigraphic section, normal and reversed directions from consecutive flows are antipolar. (2) Consistent directions were obtained from sites 30 km apart. (3) Application of the fold test indicated the magnetization was acquired before the rocks were folded. (4) The magnetizations of several pilot specimens are thermally stable up to 550°C. The stable component is probably carried by magnetite with lamellar texture, a primary feature commonly acquired by a basalt at high temperature during initial cooling of the magma. Geologic and paleomagnetic evidence indicates that the Nikolai is allochthonous to Alaska and that, together with associated formations in southern Alaska and British Columbia, it is part of a now disrupted equatorial terrane.

Les données paléomagnétiques indiquent que la vaste zone de basalte datant du début du Mésozoïque près de McCarthy, dans le centre sud de l'Alaska, provient d'une région plus au sud que sa position actuelle par rapport à l'Amérique du Nord. Les résultats obtenus pour les roches vertes de Nikolai à partir du Trias moyen ou supérieur suggèrent que ces basaltes se sont formés en dedans de 15° du paléoéquateur. Cette position est au moins à 27° (3000 km) au sud de la latitude prédite pour le Trias supérieur dans le cas de McCarthy en se basant sur les données paléomagnétiques pour le continent nord-américain. Le pôle du Nikolai, déterminé sur 50 coulées échantillonnées à 5 sites est à 2.2° N, 146.1° E (α_{95} = 4.8°). La polarité du pôle est ambigüe parce que la direction magnétique correspond à une faible inclinaison et une déclinaison à l'ouest. Donc, le Nicolai peut avoir eu son origine près de 15° N de latitude ou, alternativement, aussi au sud que 15° S de latitude. En plus de s'être déplacé vers le nord, le bloc de Nikolai a subi une rotation d'environ 90° autour d'un axe vertical. Une mesure de la fiabilité de ce pôle est fournie par les résultats favorables obtenus des tests suivants: (1) Dans une coupe stratigraphique, les directions normales et inverses pour des coulées consécutives sont antipolaires. (2) On a obtenu des directions compatibles à des sites situés à 30 km l'un de l'autre. (3) L'application du test de plissement indique que l'aimantation a été acquise avant que les roches ne soient plissées. (4) Les aimantations de plusieurs spécimens pilotes sont stables thermiquement jusqu'à 550°C. La composante stable est probablement portée par la magnétite avec une texture lamellaire, texture primaire communément acquise par un basalte à haute température durant le refroidissement initial du magma. Les observations géologiques et paléomagnétiques indiquent que le Nikolai est allochthone à l'Alaska et que l'ensemble formé avec les roches adjacentes du sud de l'Alaska et de la Colombie-Britannique fait partie d'une région équatoriale maintenant disloquée.

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Introduction

This report is based on a paleomagnetic investigation of the Nikolai Greenstone (Middle and (or) Upper Triassic) in the McCarthy quadrangle, eastern central Alaska (Fig. 1). The objective of the study was to determine a reliable paleomagnetic pole from the Nikolai basalts for comparison with the mean Triassic pole derived from rocks within the North American craton. Comparison of these poles helps to answer the following questions concerning the tectonic evolution of southern

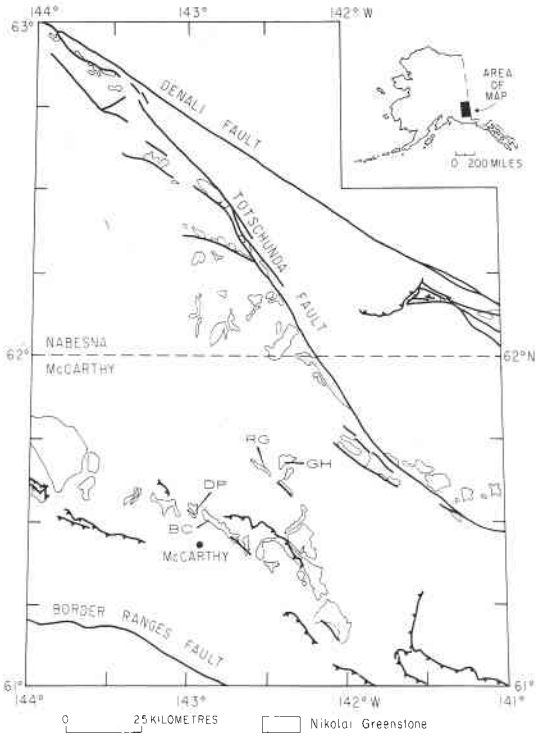


FIG. 1. Generalized map of the Nabesna (Richter 1976) and McCarthy quadrangles, Alaska, showing outcrop distribution of the Nikolai Greenstone and locations of the paleomagnetic sample sites. DP, Donoho Peak; RG, upper and lower Regal Glacier; GH, Golden Horn; BC, Bonanza Creek. Thrust faults are shown as barbed lines, barbs on upper thrust plate.

Alaska: (1) On the basis of the paleolatitude determined from the Nikolai basalts, did this terrane originate in its present position relative to cratonic North America? (2) Have the Nikolai basalts been rotated about a vertical axis since they were deposited?

The Nikolai Greenstone, named by Rohn (1900, p. 425), constitutes a thick, widespread basalt succession, largely of subaerial origin, that underlies vast tracts of south-central Alaska and extends into the Yukon Territory and probably into southeastern Alaska. It is part of a regional tectonic terrane named Wrangellia (Jones *et al.* 1977), which consists of upper Paleozoic and Triassic sedimentary and volcanic rocks. Wrangellia is coextensive with parts of the Taku–Skolai terrane of Berg *et al.* (1972). The Nikolai Greenstone is exceptionally well exposed and widely distributed along the south flank of the Wrangell Mountains in the McCarthy quadrangle. Rocks that have ages and lithologies similar to the Nikolai occur as far west as the Talkeetna Mountains (Csejtey 1976); their occurrence suggests that the belt which includes the Nikolai Greenstone extends at least 450 km west of the international border. This belt is truncated to the north by the Denali fault; basement rocks north of the fault are dominantly highly deformed Precambrian to Devonian metamorphic rocks. Richter and Jones (1973) interpreted the terrane south of the Denali fault as being floored by remnants of a late Paleozoic island arc, and they believed that pre-Albian Stage (Cretaceous) rocks of this terrane, including the Nikolai, accreted to the continental mass north of the Denali fault by thrusting.

Triassic assemblages on Baranof, Chichagof, Queen Charlotte, and Vancouver Islands are stratigraphically and paleontologically similar to the Nikolai Greenstone and its associated strata. Jones *et al.* (1977) consider these rock assemblages to be fragments of a now disrupted ancestral terrane, Wrangellia. Because Triassic and older rocks bordering Wrangellia are genetically distinct from coeval rocks within it, Wrangellia is interpreted to be allochthonous.

The paleomagnetic results reported herein provide strong evidence for the allochthonous nature of the Nikolai Greenstone. Furthermore, the paleomagnetic evidence indicates that the Nikolai originated far south of the paleolatitude predicted for McCarthy from the Upper Triassic geomagnetic pole for North America. Rather than being a locally derived belt that was thrust against the Alaskan continental margin, the Nikolai-bearing terrane appears to be an exotic crustal fragment that moved approximately 3000 km northward before being accreted to the margin of northwestern North America.

Geology

The paleomagnetic samples were collected from the rugged southern flank of the Wrangell Mountains in the McCarthy C-4, C-5, and C-6 15-min quadrangles. These quadrangles contain rocks that range from late Paleozoic (probably Pennsylvanian) to Quaternary in age and are characterized by extensive well-exposed Mesozoic sequences including abundant Nikolai Greenstone (MacKevett 1970*a*,*b*; 1972). The geologic descriptions presented here are summarized from reports by E. M. MacKevett, Jr.,

and his co-workers. Additional regional and local geologic information on the McCarthy quadrangle is given in MacKevett (1976) and in the references cited therein. The oldest exposed rocks, the upper Paleozoic Skolai Group, are interpreted as vestiges of an island arc that formed on oceanic crust. The Skolai Group generally is overlain with slight angular unconformity by a succession that sequentially includes: local thin remnants of argillite and siltstone (Middle Triassic); voluminous Nikolai Greenstone (upper Middle and (or) lower Upper Triassic); Chitistone and Nizina Limestones (Upper Triassic); McCarthy Formation (Upper Triassic and Lower Jurassic); and several marine Lower to Upper Jurassic formations. The post-Skolai Group rocks form a thick conformable sequence marked by a few disconformities and minor hiatuses and, except for the Nikolai, mainly reflect shallow-marine sedimentation. Together with the Skolai Group they constitute a coherent tectonic unit that was deformed during a Late Jurassic and Early Cretaceous regional orogeny. This tectonic unit is unconformably overlain by local Albian and younger Cretaceous marine sedimentary rocks and by the extensive subaerial Cenozoic Wrangell Lava and its subordinate associated sedimentary rocks.

The Nikolai Greenstone constitutes a thick, widespread lava field of undifferentiated tholeiitic basalt. In the McCarthy quadrangle the Nikolai is more than 3000 m thick and mainly comprises flows between 1 and 10 m thick. Generally the Nikolai overlies the Skolai Group with slight angular unconformity, but locally it disconformably overlies thin remnants of Middle Triassic siltstone and argillite. Nikolai basalts are typically amygdaloidal and, in places, have thin brecciated or rubbly flow tops. Volcanic conglomerate, as much as 70 m in aggregate thickness, locally constitutes the basal Nikolai. Most of the basalt erupted subaerially, but at a few localities in the McCarthy quadrangle some near-basal flows are pillowed, indicating subaqueous extrusion. Nikolai basalts weather to medium brown or, less commonly, greenish, yellowish, or reddish brown. They are mainly porphyritic and contain medium-grained plagioclase or augite phenocrysts in a fine-grained intergranular groundmass comprising plagioclase, augite, opaque minerals, minor relict olivine, sphene, and apatite. The secondary minerals include abundant chlorite along with hydrous iron oxides, epidote, calcite, sericite, clay minerals, prehnite, quartz, ferruginous serpentine minerals, pumpellyite, and rare zeolites. Many of these minerals occupy veinlets or amygdules. Although the alteration effects are widespread, they are rarely intense or pervasive enough to obliterate primary minerals and textures; consequently, the name 'greenstone' is a misnomer for the Nikolai basalts. The secondary mineral assemblages indicate widespread deuteric alteration and local prehnite–pumpellyite facies metamorphism.

The Nikolai is stratigraphically bracketed by rocks that contain age-diagnostic fossils, respectively the Middle Triassic pelecypod, *Daonella*, and the Late Triassic (Karnian) ammonite, *Tropites*. Hence, its age is established as late Middle and (or) early Late Triassic. Paleoclimate inferences regarding the Chitistone Limestone are relevant to the paleomagnetic interpretations. The lowermost 100 m of the Chitistone, superjacent to the Nikolai, contains abundant dolomite, relicts of evaporites, and several textures and structures indicative of intertidal–supratidal deposition and sabkha conditions (Armstrong *et al.* 1970). Modern sabkhas require arid tropical climatic regimes for their formation; hence, the lower part of the Chitistone is inferred to have formed at low latitude in an arid climate.

Paleomagnetic Sampling and Laboratory Techniques

A total of 126 oriented cores were collected from 5 sites: Golden Horn, upper Regal Glacier, lower Regal Glacier, Donoho Peak, and Bonanza Creek (Fig. 1). The cores, 2.5 cm in diameter, were drilled with portable equipment and oriented with a magnetic compass. Declination errors caused by local anomalies were removed by backsighting to reference points and applying corrections. In the laboratory the cores were cut into segments 2.28 cm long. Natural remanent magnetizations (NRM) were measured twice: shortly after the cores were collected in 1962 and again in 1976. During the 14-year period of storage, the NRM's changed only slightly, indicating that the samples acquired minor short-term viscous magnetizations. The NRM's of the more strongly magnetized samples were measured with a spinner magnetometer; the samples with intensities less

than 1×10^{-2} A/m were measured with a cryogenic magnetometer.

Secondary components of NRM were removed by alternating-field (af) demagnetization with equipment described by Doell and Cox (1967*a*). The apparatus, equipped with a four-axis tumbler, imparted spurious rotational remanent magnetization (RRM) (Wilson and Lomax 1972) to some of the specimens, so steps were taken to remove RRM. Each demagnetization run was repeated in peak fields with identical intensities but with the sense of rotation of the specimen reversed; averaging the magnetization vectors obtained from the two runs effectively removed RRM (Hillhouse 1977). Several pilot specimens from each site were progressively demagnetized in fields up to 80 mT to determine an optimum demagnetization field for the remaining specimens. The optimum field selected was that which minimized changes in magnetic direction between successive demagnetization steps and above which systematic changes in direction ceased.

Five specimens selected from three sites were thermally demagnetized up to 600°C in order to test the thermal stability of the remanence and to aid in identification of the magnetic minerals. The specimens were heated and cooled in vacuum ($\sim 10^{-5}$ Torr (13.3×10^{6} Pa)) in an ambient field of less than 50 nT. Curie temperatures of companion specimens were determined with a magnetic thermo-balance (Doell and Cox 1967*b*), which measures strong-field induced magnetization as a function of temperature (in vacuum).

Paleomagnetic Results

The paleomagnetic directions are summarized in Table 1. The site-means represent directions obtained from as many as 26 flows; in all but a few cases 2 cores were collected from each flow. Average flow directions were computed and these directions were combined in the site-mean. Alternating field demagnetization significantly reduced the dispersion of directions at each site, indicating that the NRM's were carrying magnetizations with low coercivities in addition to a more stable magnetization. Treatment at 40 mT gave the best results for all samples except those from Bonanza Creek, which required cleaning in fields up to 120 mT.

The sites fall into three groups (Fig. 2) characterized by the behavior of the remanence during af demagnetization. (1) *Golden Horn and upper Regal Glacier*: These samples showed components of low coercivity that were relatively weak compared to the stable parts of their NRM. The mean demagnetizing field (mdf), the alternating field required to reduce the magnetization to one half of its initial value, usually exceeded 40 mT and the directions were remarkably consistent in fields up to 80 mT. (2) *Lower Regal Glacier and Donoho Peak*: Low coercivity components dominated the NRM's, giving mdf's that ranged between 5 and 25 mT. Demagnetization at 40 mT improved the directional scatter considerably, but some specimens retained their secondary components, as evidenced by differences in direction of as much as 40° within individual flows. Of the 79 cores collected at these 2 sites, 25 cores were rejected because they retained magnetic directions parallel to the present field or the magnetic direction failed to converge to a stable direction during stepwise demagnetization. (3) *Bonanza Creek*: Although af demagnetization decreased the dispersion, some of the specimens retained steeply inclined components with coercivities that exceeded 120 mT. Optimum demagnetizing fields that ranged from 40 to 120 mT were selected individually. Six cores from a total of 22 cores were rejected because they apparently were overprinted with magnetization directed along the present field direction. The presence of a stable overprint is reflected in the low precision parameter (k) obtained for the Bonanza Creek site.

The thermal stability of NRM was investigated for representative specimens from Donoho Peak, lower Regal Glacier, and Bonanza Creek. In Fig. 3 the thermal demagnetization of NRM (J_n) is compared with the variation with temperature of strong-field induced magnetization (J_s). With the exception of flow 6 from the Bonanza site, the heating and cooling curves of J_s are reversible and give Curie temperatures of 570°C–580°C. These reversible curves indicate that the bulk magnetic mineral, probably a titanium-poor titanomagnetite, remains unchanged during heating, and, therefore, low-temperature oxidation products such as titanomaghemite and maghemite are not present in significant amounts. There is, however, evidence of alteration in the magnetic minerals of flow 6 (Bonanza Creek), as demonstrated by its irreversible J_s curve.

TABLE 1. Summary of paleomagnetic results from the Nikolai Greenstone, McCarthy Quadrangle, Alaska

Flow no.	N	I	D	I_c	D_c	R	k	α_{95}	θ	ϕ	H
(1) Golden Horn (Site Location: 61.69° N, 217.60° E)											
1	2	−51.1	73.9	−22.7	81.5				−6.4	320.7	40
2	2	−46.7	72.6	−18.5	79.6				−3.5	321.2	40
3	2	−48.3	70.2	−20.5	78.3				−3.9	322.9	40
4	2	−50.1	68.1	−22.6	77.5				−4.5	324.2	40
5	2	−48.7	60.8	−22.6	72.0				−2.1	328.9	40
6	1	−49.0	66.2	−25.4	73.8				−4.3	328.1	40
8	2	−51.9	61.8	−20.7	75.0				−2.5	325.8	40
9	1	−49.0	66.2	−21.8	75.8				−3.4	325.4	40
10	2	−54.7	72.2	−26.4	81.3				−8.2	321.8	40
12	1	−57.9	62.9	−30.9	76.7				−8.5	327.1	40
Site mean	10			−23.2	77.1	9.9715	316.1	2.7	−4.7	324.6	
(2) Upper Regal Glacier (Site Location: 61.66° N, 217.45° E)											
5	2	−50.7	100.5	−26.3	77.9				−6.6	324.6	40
6	2	−55.5	103.6	−31.3	76.4				−8.6	327.3	40
7	3	−69.0	82.3	−36.8	57.5				−4.0	345.1	40
Site mean	3			−31.8	71.0	2.9640	55.6	16.7	−6.5	332.3	
(3) Lower Regal Glacier (Site Location: 61.66° N, 217.45° E)											
8	3	−73.6	124.1	−48.5	65.1	—	—	—	−15.0	342.6	40
(4) Donoho Peak (Site Location: 61.58° N, 217.05° E)											
1	2	32.3	257.5	36.2	276.9				20.9	131.0	40
2	2	16.2	249.6	26.1	260.0				7.4	142.3	40
3	2	10.5	238.5	25.8	246.4				1.2	154.1	40
5	1	21.0	259.5	25.7	271.7				12.7	132.1	40
6	2	−15.4	58.4	−36.0	68.6				−7.9	335.0	40
7	2	33.4	49.8	7.9	44.4				23.6	347.4	40
8	2	8.2	40.0	−18.7	41.0				12.0	355.7	40
9	2	27.4	55.7	4.5	52.0				19.1	340.6	40
10	2	49.7	58.1	26.6	48.3				31.4	339.0	40
11	1	4.5	82.5	−9.5	83.5				−1.1	315.0	40
12	2	20.7	71.4	2.0	67.7				11.3	326.4	40
13	2	17.7	48.0	−6.4	47.0				15.9	347.6	40
14	2	−1.7	49.8	−25.1	52.6				4.6	346.2	40
15	2	8.8	46.3	−15.4	46.9				11.7	349.4	40
16	2	9.9	65.0	−10.2	65.0				7.0	331.6	40
17	2	−24.6	71.5	−40.9	85.8				−18.5	322.2	40
18	2	7.0	57.2	−15.8	58.4				7.1	338.9	40
19	2	−10.5	68.5	−29.1	75.8				−7.1	326.8	40
20	2	2.4	66.1	−17.7	68.6				1.9	330.1	40
21	2	25.5	54.1	1.5	51.1				18.1	342.1	40
22	1	−3.4	60.8	−24.9	65.1				−0.2	335.0	40
23	2	6.4	66.6	−13.8	67.8				4.1	329.9	40
24	2	−21.7	73.0	−37.7	85.8				−16.5	321.0	40
25	2	−3.3	67.0	−22.8	71.5				−1.9	328.8	40
26	2	16.7	70.1	−2.9	67.8				9.1	327.5	40
27	2	−20.5	57.7	−42.1	68.0				−11.5	337.5	40
Site mean	26			−17.6	65.4	24.1973	13.9	7.9	2.7	332.8	
(5) Bonanza Creek (Site Location: 61.51° N, 217.14° E)											
2	1	−28.6	72.1	−39.4	83.7				−16.6	323.5	60
3	2	−22.4	83.6	−29.9	92.8				−15.4	312.5	60
4	1	10.6	72.9	−1.7	71.7				7.9	323.7	60
5	2	37.7	110.5	35.3	95.6				14.4	292.8	40, 60
6	1	19.0	99.4	14.3	93.5				4.7	300.6	60
7	2	−5.1	105.9	−6.2	107.9				−11.2	292.7	60, 120
8	1	−28.8	105.7	−28.4	116.7				−25.9	290.6	60

TABLE 1 (*Concluded*)

Flow no.	N	I	D	I_c	D_c	R	k	α_{95}	θ	ϕ	H
9	2	−23.9	99.2	−26.1	108.6				−20.9	297.4	60
10	1	−21.8	83.1	−29.5	92.1				−14.8	313.0	60
11	2	−23.7	81.4	−31.8	91.2				−15.7	314.5	120, 40
Site mean	10			−15.3	95.4	9.0233	9.2	16.8	−9.6	306.2	

NOTES: Mean of site VGP's: $\theta = 6.8$, $\phi = 147.7$, $R = 4.8653$, $k = 29.7$, $\alpha_{95} = 14.3$, ($N = 5$).
Mean of flow VGP's: $\theta = 2.2$, $\phi = 146.1$, $R = 47.3177$, $k = 18.3$, $\alpha_{95} = 4.8$, ($N = 50$).
Because all flows that have easterly declinations are assumed to be of reversed polarity, the poles corresponding to these flows have been inverted in determining the mean.
N, Number of specimens per flow. I, Inclination not corrected for tilt of bedding. D, Declination not corrected for tilt of bedding. I_c, Inclination corrected for tilt of bedding. D_c, Declination corrected for tilt of bedding. R, Length of resultant vector. k, Precision parameter (Fisher 1953). α_{95}, Half angle of 95% confidence cone. θ, Latitude of virtual geomagnetic pole. ϕ, Longitude of virtual geomagnetic pole (East Longitude positive). H, Peak field used during alternating field demagnetization (mT).

Progressive thermal demagnetization of the NRM revealed two types of magnetization in four of the five test specimens: a magnetization with blocking temperature (T_B) below 250°C and a more stable magnetization with T_B between 505°C and 580°C. The single exception, flow 15 from lower Regal Glacier, exhibited T_B ranging between 100°C and 200°C but lacked a distinct blocking point at higher temperature. The remanence removed at low temperature (<250°C) is believed to be viscous remanence, (VRM) acquired by the basalt after it had cooled; the bulk of this low T_B magnetization was probably acquired during the Brunhes normal polarity epoch, although some specimens may have acquired VRM at an earlier time during burial metamorphism. The remanence that remains stable during heating to 550°C is assumed to be a thermoremanent magnetization (TRM) or possibly a high-temperature thermochemical remanent magnetization (TCRM) acquired as the basalt cooled. Because the metamorphic assemblages of the highest grade found in the Nikolai Greenstone are of the low-temperature prehnite–pumpellyite facies, it is improbable that the stable magnetization is a secondary TRM acquired during metamorphism rather than during original cooling of the basalt.

The magnetic directions obtained by partial thermal demagnetization and af demagnetization were compared (Fig. 4). For each example the results were obtained from two specimens cut from the same core. For the flows sampled at Donoho Peak, directions obtained after treatment in a 40 mT alternating field are in excellent agreement with those obtained by heating to 550°C. On the other hand, partial thermal demagnetization of the specimen from flow 15, lower Regal Glacier, revealed that the magnetization is unstable. Progressive heating failed to isolate a stable direction, although the magnetization appeared to be stable during progressive af demagnetization. With the exception of flow 8, the remaining flows from lower Regal Glacier gave directions (before correcting for tilt of the beds) that loosely cluster about the present field direction, indicating that they were remagnetized. For this reason, and because the representative sample is thermally unstable, the results obtained from eight of the flows at lower Regal Glacier were rejected. The results from flow 8 were retained because the directions were consistent after thermal and af cleaning, and the cleaned directions were significantly different from the present field direction.

Thermal demagnetization of J_n of flow 6, Bonanza Creek, revealed a large magnetization with low T_B amounting to approximately 70% of the total NRM (Fig. 3E). A stable direction was isolated by heating the specimen to 550°C. The low T_B component has very high coercivity (Fig. 4), requiring af treatment to 100 mT for its removal. The irreversible behavior of J_s and the presence of low T_B magnetization with high coercivity suggest that the magnetization is dominated by CRM created by low-temperature oxidation. Because the specimens from Bonanza Creek were given blanket treatment with af demagnetization (thermal demagnetization of all specimens was not feasible), the low T_B component probably was not removed entirely from some specimens. This is reflected in the low precision of the mean direction of the Bonanza Creek site.

As a supplement to the thermal experiments, polished sections taken from each of the five

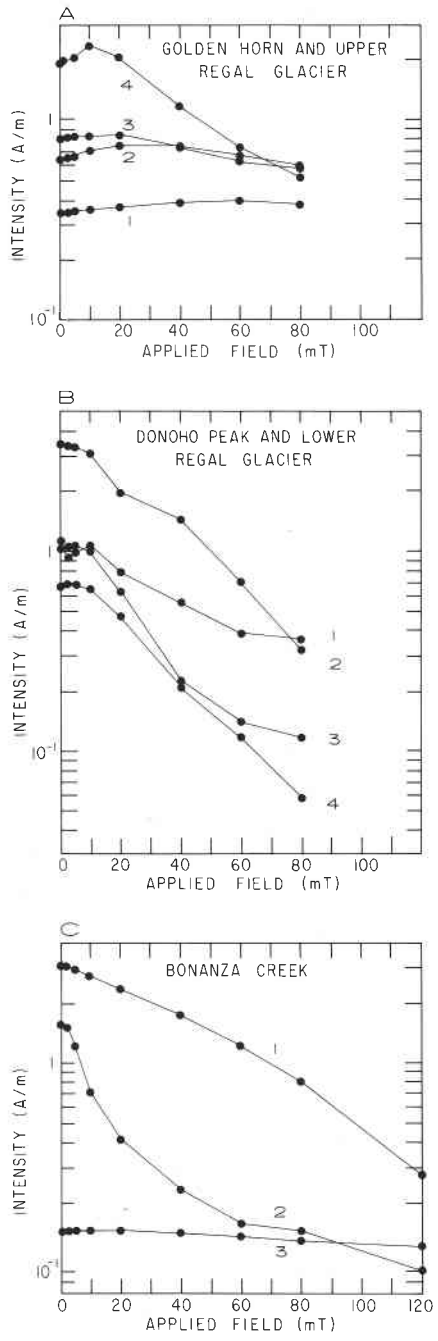


FIG. 2. Progressive af demagnetization of representative specimens of Nikolai Greenstone. (A) Golden Horn: 1, flow 5; 2, flow 3. Upper Regal Glacier: 3, flow 6; 4, flow 7. (B) Lower Regal Glacier: 1, flow 18; 2, flow 8. Donoho Peak: 3, flow 6; 4, flow 17. (C) Bonanza Creek: 1, flow 5; 2, flow 11; 3, flow 9.

test specimens were examined in reflected light. All specimens contain abundant crystals of titanium-poor titanomagnetite that exhibit a wide range of oxidation states. Many of the larger magnetite grains contain intergrowths of titanohematite and possibly sphene in fine lamellae apparently controlled by the crystal structure of the parent grains. This lamellar texture is commonly observed in fresh subaerial basalts and is believed to be due to high-temperature oxidation of primary titanomagnetite during cooling of the magma (Buddington and Lindsley 1964). However, the lamellae observed in fresh basalts typically consist of ilmenite rather than titanohematite or sphene. In the Nikolai basalts the replacement of ilmenite by sphene is probably the result of chemical alteration during burial metamorphism. Sphene lamellae associated with magnetite have been observed in prehnite–pumpellyite facies metabasalts of the Karmutsen Formation on Vancouver Island, where the replacement of ilmenite was attributed to burial metamorphism (Kuniyoshi and Liou 1976). Assuming certain physical conditions (pressure, chemical compositions, etc.), Kuniyoshi and Liou estimated that prehnite–pumpellyite metamorphism of the Karmutsen basalts occurred at temperatures between 250°C and 350°C.

Although primary ilmenite within the Nikolai has been replaced during reheating, the magnetite is probably of primary origin. Lamellar texture possibly may form in magnetite during intense metamorphism; it is unlikely, however, that the relatively low temperatures implied by the secondary-mineral assemblage in the Nikolai were sufficient to create new magnetite with lamellar intergrowths. The temperatures required for high-temperature oxidation of titanomagnetite depend on many factors, and the phenomenon has not been studied extensively in metamorphic rocks. In samples of tholeiitic basalt quenched at various depths within a Hawaiian lava lake, the formation of magnetite–ilmenite intergrowths usually occurred in the 550°C–600°C temperature range (Grommé *et al.* 1969).

In addition to the alteration mentioned above, some specimens show evidence of low-temperature oxidation, perhaps due to weathering or hydrothermal alteration. For example, flow 6 from Bonanza Creek contains a large amount of

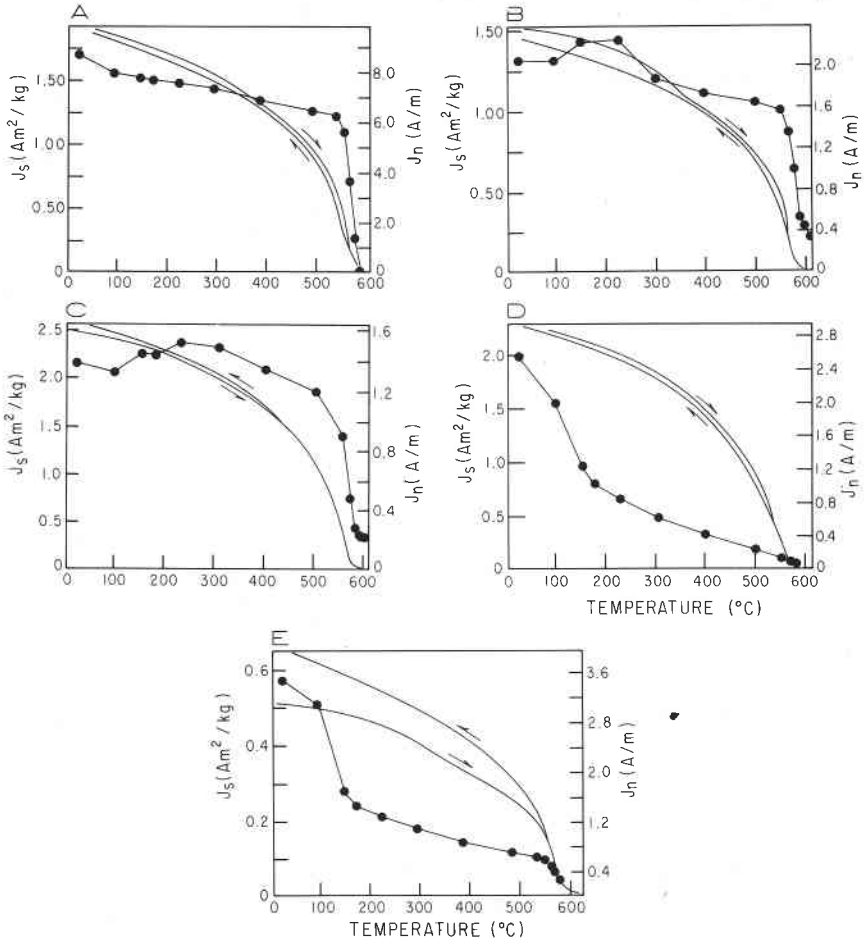


FIG. 3. Thermal behavior of strong-field induced magnetization, J_s (solid line), and NRM, J_n (dotted line), for representative specimens of Nikolai Greenstone. All samples were heated in vacuum. Arrows indicate heating and cooling curves of J_s. J_n was measured at room temperature. (A) Donoho Peak, flow 23. (B) Donoho Peak, flow 3. (C) Lower Regal Glacier, flow 8. (D) Lower Regal Glacier, flow 15. (E) Bonanza Creek, flow 6.

hematite that occurs as red pigment or as finely crystalline haloes surrounding magnetite grains. Typically, such hematite occurrences form by the oxidation of magnetite and iron-bearing silicates. Magnetic remanence carried by the secondary minerals may be responsible for the large angular dispersion of directions obtained from the Bonanza site. In contrast, specimens from Donoho Peak contain fresh magnetite that is relatively free of hematite pigment. These petrographic observations are generally consistent with the thermal behavior of J_n and J_s of the pilot specimens.

Paleomagnetic Reliability Tests

Before a paleomagnetic pole can be deemed reliable, the results upon which it is based must satisfy the following requirements: (1) The stable magnetization must have been acquired as the rocks were formed, and the rocks must be free of consistent components of secondary magnetization. (2) The results must span a time interval long enough to average out secular variation, leaving a mean magnetic direction that approximates the direction of a geocentric axial-dipole field. The first requirement can be tested either by comparing the magnetic direc-

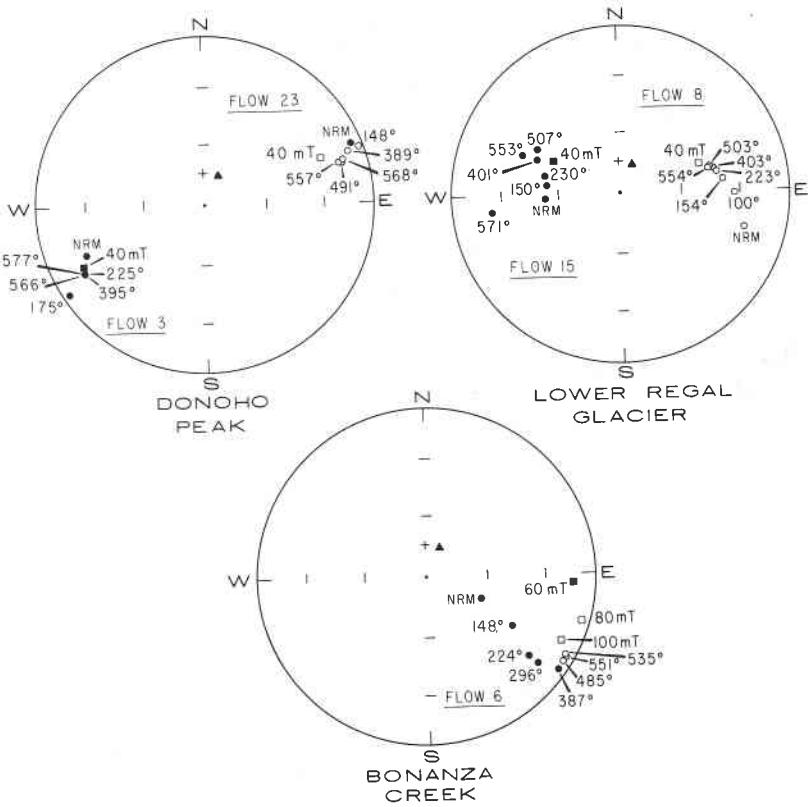


FIG. 4. A comparison of magnetic directions after progressive thermal and af demagnetization of NRM. Thermal steps are indicated by circles; Alternating field-cleaned directions are indicated by squares. Solid symbols denote positive (downward) inclinations, open symbols denote negative (upward) inclinations. (+) Axial dipole field. 1965 IGRF dipole field is indicated by a solid triangle.

tions of specimens collected from the opposite limbs of a fold or by determining whether or not normal and reversed directions are antipolar. The occurrence of coherent normal and reversed polarity zones within a stratigraphic section ensures that the second requirement is satisfied, because the periodicity of reversals generally exceeds the duration of nondipole fluctuations. The following discussion provides strong evidence that the cleaned paleomagnetic directions in Table 1 satisfy these requirements and provide the basis for a reliable Triassic paleomagnetic pole from the Nikolai Greenstone. The best evidence of primary remanence was found at Donoho Peak (Fig. 5), where a polarity reversal was recorded in a thick stratigraphic section spanning 27 flows. Four flows at the base of the section have a mean direction ($I = 29.0°$; $D = 263.4°$) that is displaced 156° from the mean direction ($I = -15.3°$; $D = 62.3°$) of the overlying flows. These two groups are antipolar at the 95% confidence level, indicating that the directions are free of consistent secondary components. The recording of a geomagnetic reversal in a stratigraphic succession provides strong evidence that the magnetization was acquired during initial cooling of the basalts, not during some later episode of thermal or chemical magnetization. Also, the presence of a reversal is evidence of a field produced by a geocentric dipole, ruling out the possibility that the Nikolai basalts were magnetized during a non-dipole excursion of the field.

In the McCarthy quadrangle the Nikolai Greenstone has been gently folded about a horizontal axis trending roughly northwest; at the northern paleomagnetic sites the flows dip

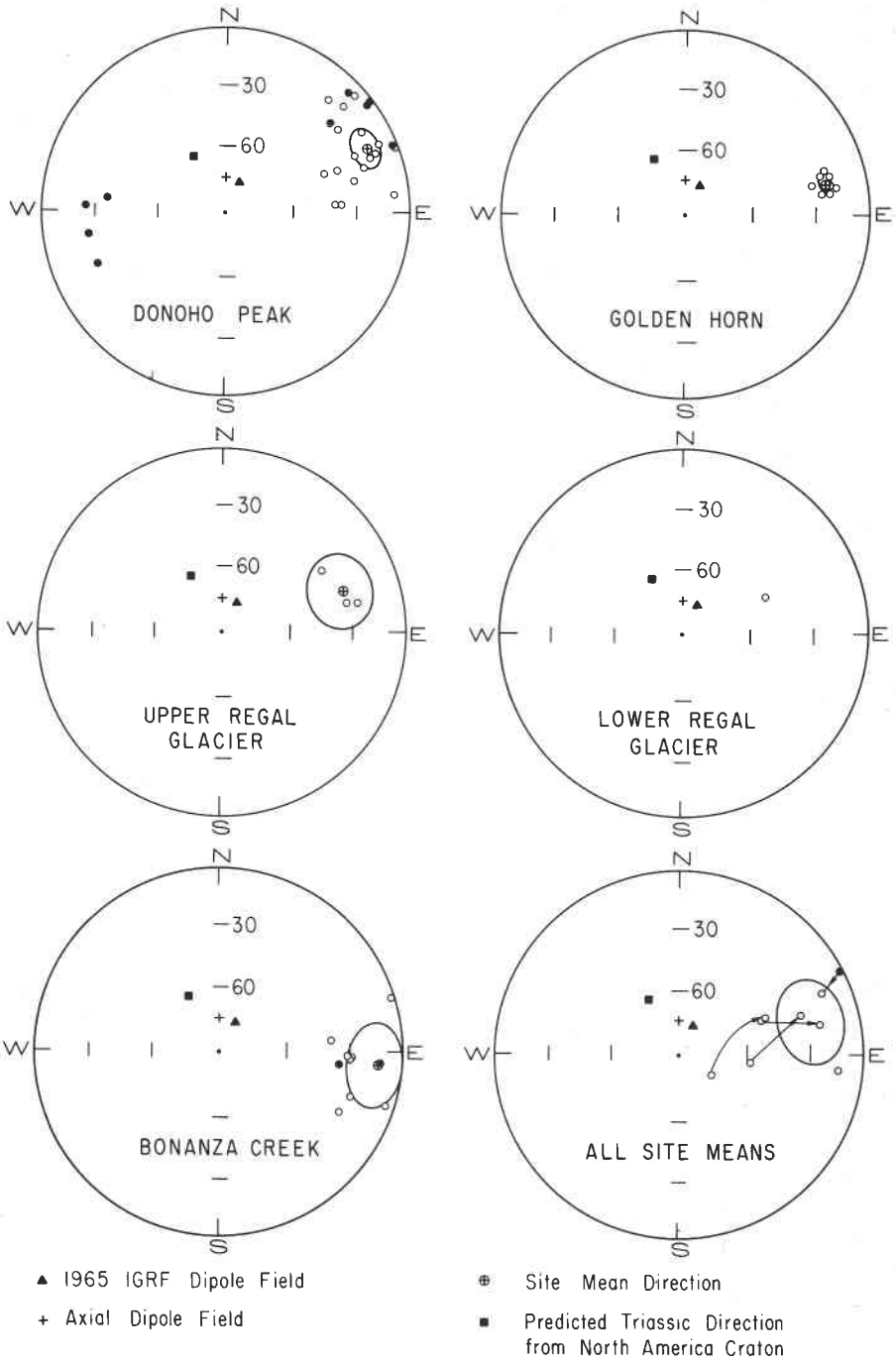


FIG. 5. Summary of alternating field-cleaned paleomagnetic directions of the Nikolai Greenstone, grouped by sites. Arrows in diagram labeled 'All site means' indicate the change from *in situ* directions to directions corrected for tilt of the bedding. In the other diagrams all directions have been corrected for tilt of the bedding. Dots indicate directions with positive (downward) inclinations; circles indicate negative (upward) inclinations. Ninety-five percent confidence limits of the means are shown.

30–45° southward, at the southern sites, 20–30° northward. A comparison was made of the scatter in site means (Donoho Peak, Golden Horn, upper and lower Regal Glacier) before and after the bedding corrections were made (Fig. 5). The results from Bonanza Creek were omitted because the internal scatter exceeded the angle of the tilt correction. The Fisher precision parameter (k) increased from 5.1, the *in situ* value, to 32.1, the corrected value, meaning that application of the bedding correction reduced the angular standard deviation of the mean by a factor of 2.5. Therefore, the magnetization was acquired before the beds were folded, and this fact provides further evidence that the magnetization is primary.

The amount of directional scatter varies between sites (Fig. 5), but the directions generally fall into two groups with opposite polarities having east–west declinations and low inclinations. The mean direction obtained from Donoho Peak is probably the most reliable, because the sampled stratigraphic section represents the greatest number of flows and includes a geomagnetic reversal. Similar mean directions were obtained from the Golden Horn and upper Regal Glacier sites, located approximately 30 km northeast of Donoho Peak. The consistent results show that despite the presence of many faults in the area, the three sites lie within a coherent block that has not been disrupted by internal rotations. This is not true for the Bonanza Creek site, which has a mean direction that is displaced 30° east of the main group of means. The rotation may not be significant because the Bonanza cores apparently retain magnetization with low T_B that could not be removed by af demagnetization; thus the mean direction is associated with a large uncertainty.

Although some flows retain small random components of secondary magnetization, the paleomagnetic results are generally consistent, include normal and reversed polarities, and were obtained from sites distributed over a large distance. On the basis of these results, it is therefore valid to determine the mean magnetic direction and paleolatitude of the Nikolai Greenstone and to use the result in discussions of the formation's tectonic history. In calculating the overall mean, the two-tier analysis of Watson and Irving (1957) was not used because the necessary assumption that the within-site dispersion be constant is not fulfilled here. Instead, each measurement was given equal weight and the formation mean was calculated by repeated use of Fisher's method (Fisher 1953). Combining the five site means gave an overall mean direction of ($D = 75.5°$, $I = -27.7°$) with a 95% confidence radius of 16.9° ($k = 21.5$). In this calculation the westerly directions from Donoho Peak were inverted through the origin. The analysis tends to overestimate the true dispersion, because the sites having greater uncertainties, lower Regal Glacier and Bonanza Creek, were weighted equally against the more reliable sites. Therefore, the overall mean was also calculated by combining the 50 individual flows giving ($D = 74.0°$, $I = -20.3°$) with a 95% confidence radius of 5.9° ($k = 12.5$). The best estimate of α_{95} lies between 5.9° and 16.9°.

If we assume that an axial-dipole field existed, the paleolatitude (θ) of the Nikolai basalt can be determined from the mean magnetic inclination (I) by the relation $\theta = \tan^{-1}((\tan I)/2)$. In this calculation it is assumed that the mean inclination is produced by a normally polarized (north-seeking) field. However, the negative inclination found in the Nikolai basalt can be interpreted in two ways: (1) the rock originated in the southern hemisphere during a normal polarity epoch, or (2) the rock originated in the northern hemisphere during a reversed polarity epoch. Because the declinations in the Nikolai are nearly east–west, probably owing to large-scale rotation of the Nikolai block, this ambiguity cannot be resolved. Using the Late Triassic paleomagnetic pole for stable North America as a reference, I have assigned normal polarity to the magnetic directions from the Nikolai that have westerly declinations. Of the two possibilities, this interpretation requires a smaller rotation to make the Nikolai pole coincide with the North American pole. On the basis of the more conservative mean direction ($D = 255.5°$, $I = 27.7°$, $\alpha_{95} = 16.9°$) calculated from the five site means, the paleolatitude of the Nikolai basalt is 15° N, with 95% confidence limits of 26° N and 5° N. As mentioned previously, the true confidence interval is probably smaller. Of course, the assignment of polarities might be incorrect, in which case the paleolatitude would be 15° S.

Conversion of each flow direction to its virtual geomagnetic pole (VGP) gave a mean pole for the Nikolai at (2.2° N, 146.1° E, α_{95} =

TABLE 2. Late Triassic paleomagnetic poles from North America

Locality	K–Ar age (Ma)	θ	ϕ	α_{95}	Reference
1. Newark Group, New Jersey	202	63° N	108° E	3°, 4°	Opdyke (1961)
2. Lavas, Massachusetts	205–180	55° N	88° E	6°, 11°	Irving and Banks (1961)
3. Diabase, Nova Scotia	197	69° N	98° E	4°, 6°	Larochelle and Wanless (1966)
4. Diabase, Pennsylvania		62° N	105° E	2°, 3°	Beck (1965)
5. North Mountain basalt, Nova Scotia	200 ± 10	73° N	104° E	5°, 7°	Carmichael and Palmer (1968)
6. Igneous Rock, Connecticut Valley	193 ± 6	65° N	87° E	7°, 12°	de Boer (1968)
7. Kayenta Formation, Utah		62° N	74° E	4°, 7°	Steiner and Helsley (1974)
8. Chinle Formation, New Mexico		58° N	79° E	2°, 4°	Reeve and Helsley (1972)

NOTES: Mean pole for North America: $N = 8$, $\theta = 64°$ N, $\phi = 92°$ E, $\alpha_{95} = 5.5°$.
θ = latitude of mean VGP. ϕ = longitude of mean VGP. α_{95} = minor and major axes of 95% confidence ellipse.

4.8°) in the western Pacific Ocean. The westerly directions were assigned normal polarity; the remaining directions were inverted through the origin.

Displacement of the Nikolai Greenstone: Allochthonous Crust in Southern Alaska

Paleomagnetic poles of Late Triassic age from the tectonically stable part of North America (Table 2) provide a reference pole for comparison with the displaced Nikolai pole (Fig. 6). The sites include igneous rocks from the eastern United States and sedimentary red beds from the southwest. From the available data it is not possible to determine a geomagnetic pole that is exactly coeval with the Nikolai Greenstone, and some of the units included in Table 2 are probably too young. Between Early Jurassic and Late Triassic time, however, northward drift of North America did not exceed 12° (Van der Voo and French 1974); in view of the scatter in the paleomagnetic data, Table 2 provides an adequate reference pole. As presented here, the Late Triassic pole for North America is at (64° N, 92° E) ($N = 8, \alpha_{95} = 5.5°$).

If Alaska has remained fixed with respect to North America, then the Late Triassic paleolatitude of McCarthy should be 42.5° N, considerably farther north than the paleolatitude determined from the Nikolai Greenstone. At the 95% confidence level, the Nikolai formed between 11° and 43° south of the position predicted from the North American reference pole. The evidence suggests that the Nikolai did not originate in its present position with respect to North America; instead it was deposited at low latitude and was carried northward roughly 3000 km relative to the North American craton. The observed difference in paleolatitude can be explained only by rotation that includes a large northward component.

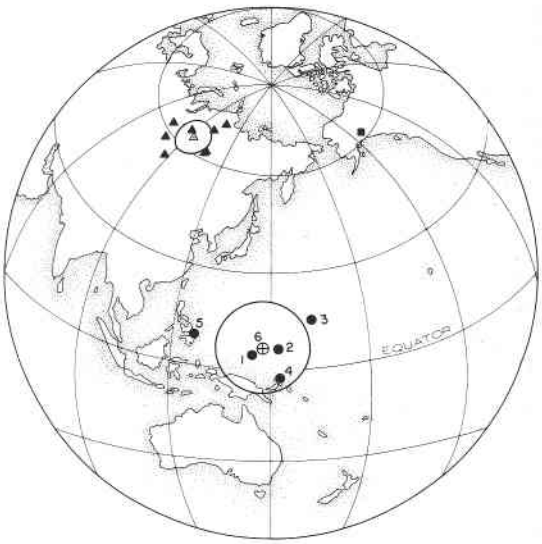


FIG. 6. Virtual geomagnetic poles (dots) corresponding to the mean directions determined from five sites within the Nikolai Greenstone. (1) Golden Horn. (2) Upper Regal Glacier. (3) Lower Regal Glacier. (4) Donoho Peak. (5) Bonanza Creek. (6) Mean of 1–5 with α_{95} confidence circle. For comparison, VGP's (solid triangles) and the mean (dotted triangle) Triassic North American reference pole, listed in Table 2, are shown. Location of study area is indicated by a solid square.

A problem that remains to be solved concerns the location of the Nikolai source area. Although the paleomagnetic results provide the paleolatitude of the source area, they do not define its paleomeridian with respect to North America. The relative movement between the Nikolai and North America can be described as a single rotation about a pivot point that lies along the great circle equidistant from the Nikolai and Triassic North American geo-

magnetic poles. If an older pair of coeval poles were available from North America and the Nikolai basement, then the pivot point and the amount of rotation would be defined. In this way, the past relative positions of two blocks could be uniquely determined, provided that no relative movement between the sites had occurred prior to the Triassic. Unfortunately a pre-Triassic pole has not been determined from the McCarthy area.

Some constraints on the location of the Nikolai source area are provided by the Triassic distribution of land masses, reconstructed from paleomagnetic data (Van der Voo and French 1974). As shown in Fig. 7, the Nikolai basalts could have originated along 15° N in the ancient Pacific Ocean anywhere between the west coast of North America and the east coast of Asia. If we assume that the Nikolai has moved the shortest distance possible, then it originated off the southern coast of California, probably as a plateau basalt field that superseded the island arc represented by the Skolai Group. The pivot point would be located in the Pacific, and the rotation required to bring the North American and Nikolai paleomagnetic poles into their present positions would be approximately 70° counterclockwise. Because the continental margin of North America would interfere with this hypothetical rotation, the motion must be broken into two rotations: northward translation and counterclockwise rotation about a vertical axis centered in the Nikolai block. The present position of the Nikolai block can be explained by right-lateral motion that carried the Nikolai northward along the northeast margin of the Pacific Ocean. The block then underwent counterclockwise rotation as it collided with the margin of Alaska. This hypothesis utilizes many assumptions, and it is only one of many possible reconstructions that describe the origin and movement of the Nikolai basalts. Because an ambiguity remains in the polarity determination, the sense of rotation of the Nikolai block cannot be determined with certainty. For this reason, it is not appropriate to use these results to prove or disprove hypotheses concerning oroclinal bending of Alaska (Carey 1958; Freeland and Dietz 1973).

It remains to be demonstrated whether or not, on geologic grounds, this large-scale northward displacement is possible. In speculations concerning Mesozoic plate interactions, geologists have emphasized the role of east–west subduction in forming the Cordilleran orogen (Rogers *et al.* 1974; Monger *et al.* 1972; Burchfiel and Davis 1972; Dickinson 1976). However, the possibility of large-scale transform motion between North America and the oceanic plates to the west in late Mesozoic time finds some support from paleontologic data (Monger and Ross 1971), paleomagnetic data (Irving and Yole 1972), and the apparent truncation of Mesozoic belts in California (Hamilton 1969; Schweickert 1976).

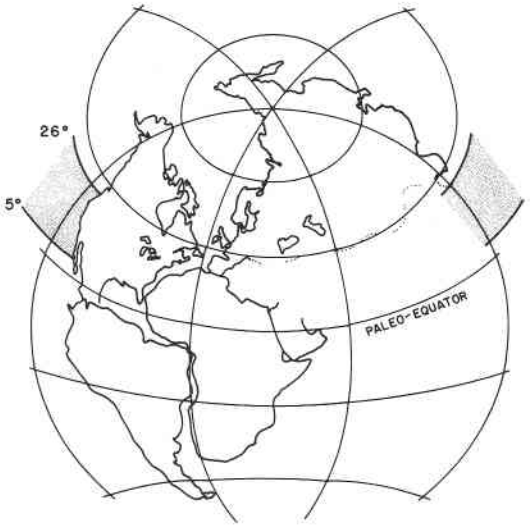


FIG. 7. Position of continents bordering the Atlantic Ocean in Upper Triassic time (equidistant projection). Paleolatitudes were inferred from mean paleomagnetic pole calculated by Van der Voo and French (1974). If the eastward paleomagnetic directions shown in Fig. 5 have reversed polarity, the Nikolai Greenstone probably originated at a paleolatitude lying within the shaded region. Boundaries of the shaded region (5° N–26° N) correspond to the 95% confidence limits of the mean Nikolai direction, calculated from the 5 site means. The solid triangle denotes the present location of Nikolai Greenstone with respect to North America.

Because the polarity of Nikolai cannot be determined unambiguously, it is also possible that the Nikolai Greenstone originated in the southern hemisphere, between 26° S and 5° S.

Conclusions

The paleomagnetic pole (2.2° N, 146.1° E, $\alpha_{95} = 14.8°$) determined from the Nikolai Greenstone is displaced 73° from the Late Triassic reference pole for North America. The results suggest that in addition to a possible

counterclockwise rotation, the Nikolai Greenstone was displaced roughly 3000 km northward before reaching its present position relative to North America. Because the polarity of Nikolai cannot be determined unambiguously, it is also possible that the Nikolai basalts originated in the southern hemisphere. If so, the northward displacement would be 57° or approximately 6300 km, and the rotation would be clockwise.

A measure of the reliability of the paleomagnetic directions determined from the Nikolai is provided by favorable results from the following tests: (1) Within one stratigraphic section, normal and reversed magnetozones give antipolar directions, indicating that the results are not biased by secondary remanence and that the rocks did not record a short interval of anomalous field behavior. (2) Consistent directions were obtained from sites 30 km apart. (3) Application of the fold test showed that the magnetization was acquired before the rocks were folded. (4) The stable magnetizations of several pilot specimens have blocking temperatures between 500°C and 575°C, probably acquired as TRM or TCRM during initial cooling of the basalts. The lamellar texture of magnetite observed in these specimens is a common feature of stably magnetized igneous rocks (Larson *et al.* 1969; Strangway *et al.* 1968).

The evidence presented here suggests that the Nikolai Greenstone was once detached from northwestern North America; however, our understanding of the allochthon's boundaries and the timing of its accretion to Alaska is limited by the lack of paleomagnetic data from adjacent rocks. On the basis of geologic and paleontologic evidence from the coast of British Columbia and southeastern Alaska, Jones *et al.* (1977), have speculated that the Nikolai basalts are only a fragment of a now disrupted ancestral terrane (Wrangellia) that included the Triassic volcanic rocks of Vancouver Island, Queen Charlotte Island, Chichagof Island, and Baranof Island. This hypothesis profoundly affects the interpretation of Cordilleran geology and must therefore be carefully weighed against other interpretations derived from the same geologic evidence. It is important to note that without exception, 'Wrangellian' Triassic rocks in south-central Alaska and Vancouver Island (Irving and Yole 1972) and Jurassic sedimentary rocks in the western Alaska Range (Packer and Stone 1974) give paleomagnetic results that support an exotic, southerly origin for these rocks.

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